Aditya Attrey

adityaattrey@vt.edu +1(919)-744-8088 www.linkedin.com/in/aditya-attrey
Website: adityaattrey.com

Summary

Driven Honors Aerospace Engineering student at Virginia Tech specializing in spacecraft propulsion and avionics. Combines a strong foundation in mechanical systems and computational analysis with hands-on expertise in rapid prototyping, hardware design, and microcontroller integration (ESP32). Proven leader with experience guiding robotics teams and leveraging CAD tools (Fusion360) and programming (MATLAB, C++) to solve complex problems. Enthusiastic about applying analytical problem-solving and multidisciplinary technical skills to advance aerospace manufacturing.

Education

Virginia Polytechnic Institute and State University (Virginia Tech) — Blacksburg, VA
Bachelor of Science in Aerospace Engineering, Minor in Physics
Expected Graduation: May 2028
GPA: 3.4 / 4.0

Triangle Match and Science High School — Apex, NC
High School Diploma, May 2024
GPA: 3.8 / 4.0
Honors: AP Scholar with Distinction
Activities: Robotics Club

Skills

- **CAD & Design: SolidWorks, Fusion 360, SolidWorks, 2D/3D CAD modeling, engineering drawings, design for manufacturability (DFM)**
- **Programming & Software: MATLAB, C++, Python, Java**
- **Analysis: Fluid dynamics, cavitation analysis, structural mechanics, data analysis**
- **Hardware & Prototyping: ESP32 microcontrollers, circuit design, rapid prototyping, hardware integration, soldering and laser cutting.**
- **Engineering Methods: Experimental testing, design validation, iterative design, material selection & trade studies.**

Experience

Undergraduate Research **Feb 2025 -- Present**

- **Conduct cavitation research under Dr. Delgosha, analyzing fluid behavior under varying flow conditions**
- **Designed and modified experimental geometries in SolidWorks to control flow rate and cavitation effects**
- **Collaborated with a research team to assemble experimental test rig for controlled fluid testing**
- **Performed MATLAB-based analysis of particle flow and cavitation dynamics across varying parameters**
- **Interpreted experimental data to evaluate system performance and support design insights**

Robo-Grinder VT **Sep 2024 - April 2025**

Structural Design Co-Lead

- Designed structural components using CAD with focus on strength-to-weight optimization and design for manufacturability (DFM)
- Conducted material trade studies on aluminum alloys and carbon fiber composites to inform performance and weight-critical design decisions
- Produced detailed engineering drawings, bills of materials (BOM), and documentation in accordance with engineering standards
- Collaborated with cross-functional sub teams to integrate mechanical components into the overall system architecture
- Iterated designs through prototyping and testing to improve structural performance and reliability

FTC Robotics Club **Sep 2020 - April 2024**

Hardware Design Lead

- Directed collaborative mechanical design efforts for the competition robot, including launcher mechanisms, scissor-lift subsystems, and an all-wheel-drive (AWD) drive system.
- Managed the full design lifecycle from CAD modeling to rapid prototyping and testing, iteratively refining components to optimize performance and structural reliability.
- Organized structured CAD (SolidWorks/Fusion 360) and fabrication training for teammates, fostering a culture of safety and engineering best practices while actively procuring team sponsorships.

PERSONAL PROJECTS

2-DOF Rocket Trajectory & Propulsion Simulator

- Developed a MATLAB computational model to simulate a 2-DOF rocket flight, calculating maximum altitude and roll dynamics.
- Engineered algorithms to determine optimal fuel requirements and nozzle geometry, demonstrating practical application of propulsion and fluid mechanics.

Custom Motorized Bicycle

- Designed, engineered, and assembled a fully functional motorized bicycle completely from scratch, applying hands-on mechanical fabrication and troubleshooting skills

Automated Wireless Irrigation System (Ongoing)

- Designed and implemented an ESP32-based wireless soil sensing system, integrating custom hardware and microcontrollers for automated plant care.

TikTok-Style News Feed Platform

- Developed and deployed a dynamic, full-stack news feed platform tailored for short-form content consumption.
- Designed the underlying system architecture to successfully support and serve an active user base of 100+ individuals.

Large Language Model (LLM) Fine-Tuning

- Fine-tuned a large language model (LLM) by leveraging LoRA (Low-Rank Adaptation) techniques to ensure efficient training.
- Applied structured machine learning workflows and mathematical optimization methods to evaluate and enhance model performance.

Hobbies / Interests

- Playing guitar
- DIY electronics
- Hiking
- Watchmaking